UNIT - III : BOREHOLE LOGGING

Logging methods | Groundwater observations | W.T. fluctuation | Soil profiles | Exploration report

> Borehole log = the written + graphical record of what the drill finds at every depth. It is the most important output of exploration - all design decisions and later contract claims are based on it.

1. Logging of boreholes

Logging means systematic recording of soil/rock encountered, the drilling operation, samples taken, tests performed and ground water observations. It is carried out by a trained geotechnical engineer / geologist at the rig - not from memory at the office.

1.1 What is recorded in a log?

Group	Items recorded
Header	Project, client, location, borehole number, co-ordinates, R.L. of ground, date, type of rig, diameter, driller, logged by
Strata	Depth of top & bottom of each layer, thickness, soil description (colour, consistency/density, grain size, plasticity, moisture, smell, organic content, structure), graphic symbol, IS/USCS group symbol
Sampling & tests	Depth and type of sample (D = disturbed, U = undisturbed, S = SPT, C = core), SPT N-value (15 cm blows each), vane, PMT, etc.
Drilling record	Method, casing depth, drilling fluid loss, rate of penetration, hard layers, boulders, cavities, drilling difficulties, time taken
Water	Depth of water first met, rest water level after 24 h, casing water level, artesian flow/seepage
Rock core	Core run, TCR, RCR, RQD, weathering grade, discontinuities, rock type

1.2 Soil description - the standard order

Write in this order (IS:1498 / BS 5930): Consistency or density -> colour -> secondary constituents -> PRIMARY SOIL -> particle shape / structure -> origin/other remarks. Example: "Stiff, brown, slightly sandy, silty CLAY with calcareous nodules (CI), residual."

Coarse soil	N (SPT)	Density term
Sand/gravel	0 - 4	Very loose
	4 - 10	Loose
	10 - 30	Medium dense
	30 - 50	Dense
	> 50	Very dense

Fine soil	N (SPT)	qu (kPa)	Consistency
Clay/silt	< 2	< 25	Very soft
	2 - 4	25 - 50	Soft
	4 - 8	50 - 100	Medium
	8 - 15	100 - 200	Stiff
	15 - 30	200 - 400	Very stiff
	> 30	> 400	Hard

Stages of a subsurface investigation programme

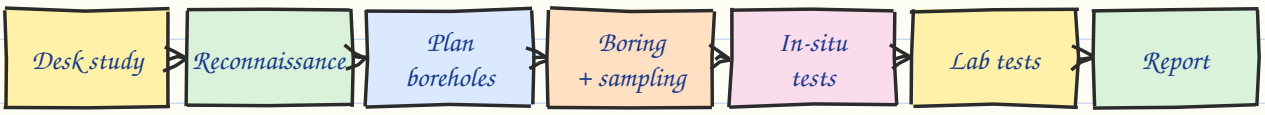


Fig 3.0 Programme of investigation

1.5 Tabular log (field sheet example)

Depth (m)	Description	Sample	SPT N (15-15-15 cm)	W.T.
0.0 - 0.5	Filled soil, brown, loose, with brick bits	D1 @ 0.5	-	-
0.5 - 2.0	Stiff brown silty clay (CI)	U1 @ 1.5	3-4-5 (N = 9)	-

Depth (m)	Description	Sample	SPT N (15-15-15 cm)	W.T.
2.0 - 4.5	Medium dense yellowish brown fine to medium sand (SP)	S2 @ 3.0	8-10-12 (N = 22)	3.2 m (24 h)
4.5 - 7.0	Dense gravelly sand with cobbles, GW	S3 @ 6.0	18-25-30 (N = 55)	-
7.0 - 8.0	Highly weathered granite (W4)	C1 (core run 1.0 m)	Refusal N > 100	-
8.0 - 11.0	Moderately weathered granite, TCR 85 %, RQD 55 %	C2	-	-

Note on SPT recording: The first 15 cm is the seating drive; N = blows for the next 30 cm (second + third 15 cm). If N > 50 for less than 30 cm penetration, record "Refusal, N > 50 for x cm" and stop.

1.6 Common mistakes and good logging practice

- Describing the soil only from the washed cuttings - always see the actual sample.
- Not giving the depth reference (ground level, casing top, platform).
- Writing "clay" for a silt or a clayey sand - do the thread test, dilatancy, dry strength, and wet-feel.
- Missing thin sand/silt partings in clay - drainage conditions depend on them.
- Giving N values without noting the hammer type, casing, and water level.
- Skipping the time and date of W.T. observation.

1.3 Graphical log (typical)

Depth (m)	Symbol	Description of strata	SPT N	Sample type / No.	W.T.
0.9		Brown silty clay, soft			
2.2		Grey clay, medium	7	SPT / D3	
3.6		Fine sand, loose	12	SPT / D6	
5.0		Medium sand, dense	31	SPT / D9	

Project: _____
BH No: 3
RL: 100.00 m
WT: 1.0 m
Date: _____

Fig 3.1 Format of a borehole log

1.4 Conventional hatch symbols

Soil / rock	Symbol (draw like this)
Gravel	Open circles (o o o) irregular
Sand	Fine dots (. . . .)
Silt	Short broken horizontal lines (- - -)
Clay	Continuous horizontal / diagonal lines
Peat / organic	Wavy lines with small vertical ticks
Rock	Bricks / zig-zag (V V V)
Filling	Cross-hatched triangle

2. Logging methods

Method	Description	Remarks
Visual logging of cuttings / samples	Strata identified from auger cuttings, washed material, SPT split-spoon sample, tube sample	Standard method; accuracy depends on skill
Drilling-time / penetration rate log	Rate of advance or drilling pressure plotted with depth (change in rate = change of strata)	Used in percussion/rotary where samples are scarce
Core logging	Rock cores laid out in core boxes: TCR, SCR, RQD, fractures, weathering	For rock and hard soil
Continuous sampling	Continuous tube sampling (e.g. Delft, Sonic) records every layer	Expensive, used for critical projects

Method	Description	Remarks
Geophysical borehole logging	Probes lowered in hole: natural gamma, resistivity, SP, density, caliper, sonic, televiewer	Good for correlation and for holes where core was lost
In-situ test logs	Cone resistance and friction ratio (CPT), N vs depth (SPT)	Continuous profile of soil stiffness

Rock core terms: TCR = Total core recovery = (length of core recovered / length of run) x 100. SCR = Solid core recovery (pieces >= full dia. of core). RQD = sum of core pieces longer than 10 cm / length of run x 100. RQD: 0-25 very poor, 25-50 poor, 50-75 fair, 75-90 good, 90-100 excellent.

Example: In a 1.5 m core run the pieces > 10 cm are 20, 15, 30, 12, 18 cm (sum = 95 cm). Recovery 1.2 m. TCR = 1.2/1.5 = 80 %. RQD = 95/150 x 100 = 63 % (fair rock).

2.1 Logging of rock cores (IS:4464)

Weathering grade	Term	Description
W1	Fresh	No visible sign of weathering
W2	Slightly weathered	Discoloration along joints; rock strong
W3	Moderately weathered	Less than half rock decomposed; core stones present
W4	Highly weathered	More than half decomposed to soil; rock fragments
W5	Completely weathered	All rock changed to soil, original structure visible
W6	Residual soil	Original structure destroyed

Joint spacing	Term
> 2 m	Very wide
0.6 - 2 m	Wide
0.2 - 0.6 m	Moderate
60 - 200 mm	Close

Joint spacing	Term
< 60 mm	Very close

Also record: rock name (from mineral/grain), colour, strength (field test: hammer blow - very weak: crumbles; weak: scratched with knife; strong: needs several blows), fracture frequency per metre, dip of joints (degrees from horizontal), infilling of joints (clay, calcite), and water loss.

2.2 Geophysical (wire-line) logging in boreholes

Log	Measures	Used to find
Natural gamma	Radioactivity of strata	Clay / shale (high) vs sand (low) - lithology
Electrical resistivity	Resistance of formation + fluid	Saturation, water quality, sand/clay
Spontaneous potential (SP)	Natural electric potential	Permeable zones
Caliper	Hole diameter	Caving zones, fractures
Sonic (velocity)	P-wave velocity	Rock quality, porosity, fractures
Density/neutron	Bulk density, moisture	Porosity of strata
Televiewer / camera	Image of borehole wall	Joints, dip & strike, cavities

3. Ground water observations

Water table (W.T.) or phreatic surface is the level at which the pore pressure is atmospheric. Its position changes with season, rainfall, pumping and tides, so it must be observed carefully and repeatedly during exploration.

3.1 How to observe the W.T. in a borehole

- Record the depth at which water is first encountered (moist -> wet sample, drilling fluid loss or gain) during advance.
- Stop drilling for the day and note the water level in the hole at the end of shift and again next morning before starting (after 24 h) - this is the "rest" or stabilised W.T. For clay the time is longer (even 1 - 2 weeks).

- In sandy soils the water in a hole comes into equilibrium within minutes to hours; in clays after days.
- Remove casing or bail the hole and watch the recovery; use a dip meter (water level indicator) or a tape with a probe sounding.
- For long term monitoring install an observation well / standpipe or a piezometer (Unit V).
- Record any artesian pressure, perched water table, and whether the water is flowing from fissures or from sand lenses.

3.2 Types of ground water

Type	Meaning
Unconfined (phreatic) aquifer	W.T. free; rises & falls with recharge. Pore pressure = $\gamma_w \times$ depth below W.T.
Perched water table	A small body of water held above the main W.T. by a thin impermeable lens
Confined / artesian aquifer	Water under pressure between two impervious layers; piezometric level may be above ground
Capillary water	Held above W.T. by surface tension: $h_c = 0.3/d$ (cm), clays can rise several metres, sands 0.1 - 1 m

3.3 Water table fluctuations and their effects

W.T. changes due to: rainfall / monsoon recharge, evaporation, pumping of wells, tides, reservoir level, irrigation canals, leaking pipes, temperature and barometric changes. Annual variation of 1 - 3 m is common in India; foundations are always designed for the highest likely W.T.

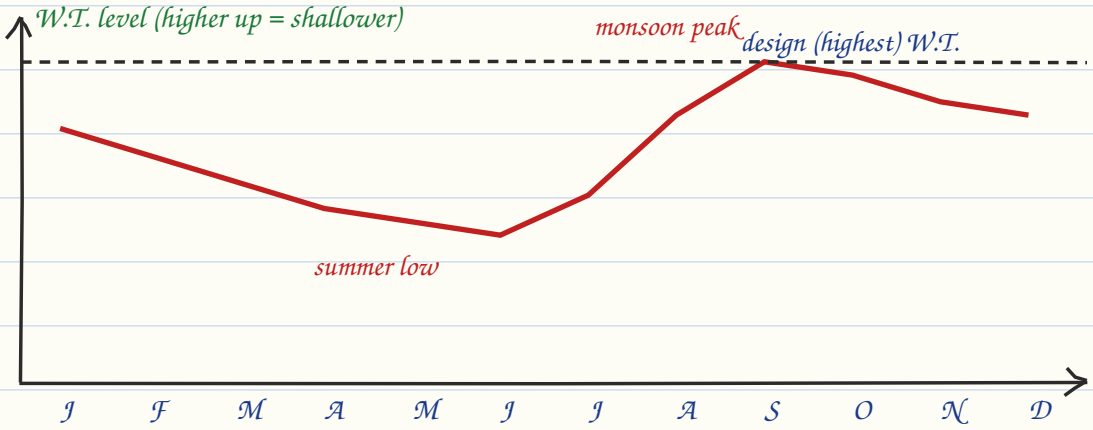
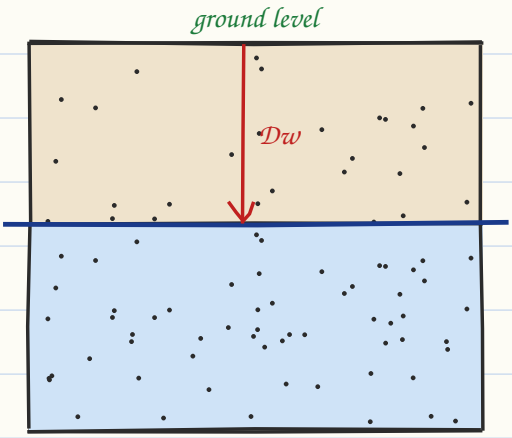


Fig 3.2a Typical seasonal fluctuation of ground water table



Effect of W.T. rising to ground:

- effective stress $s' = s - u$ decreases
- bearing capacity drops by upto 50 %
- settlement and heave increases
- buoyancy / uplift on basements
- seepage, quick sand, piping
- dewatering cost rises

Fig 3.2 Effect of a rise in W.T.

Effect	Explanation
Effective stress	Rise of W.T. reduces effective stress: $s' = s - u$. Strength and bearing capacity fall.
Bearing capacity	Terms $\gamma D_f N_q$ and $0.5 \gamma B N_\gamma$ use submerged unit weight $\gamma' = \gamma_{sat} - \gamma_w$ (about half); capacity reduces up to 50 %. Correction factor $R_{w1}, R_{w2} = 0.5 + 0.5 D_w/D_f$.
Settlement	Consolidation settlement of clays increases when W.T. falls (effective stress increases) - e.g. due to pumping: land subsidence
Swelling / heave	Wetting of expansive soils (black cotton) with rising W.T.
Uplift / buoyancy	Basements, tanks, and pile caps floating; need tension piles/ weight
Excavation problems	Seepage into excavations, quick sand, piping, loosening - need well-point dewatering
Corrosion & chemical attack	Sulphates/chlorides in ground water attack concrete & steel

Effect	Explanation
Liquefaction potential	High W.T. in loose sand is a main factor in earthquake liquefaction

SPT correction for water: in saturated fine/silty dense sands (N > 15) apply N_corrected = 15 + 0.5 (N_observed - 15) (Terzaghi & Peck).

3.4 Effect of W.T. on stress - worked example

Problem. Soil: gdry = 16 kN/m3 above W.T., gsat = 19 kN/m3. Find vertical effective stress at 6 m depth when (a) W.T. at 2 m, (b) W.T. rises to ground level. (gw = 9.81)

(a) total s = 16 x 2 + 19 x 4 = 108 kPa; u = 9.81 x 4 = 39.2 kPa; s' = 68.8 kPa

(b) s = 19 x 6 = 114 kPa; u = 9.81 x 6 = 58.9 kPa; s' = 55.1 kPa

Rise of W.T. reduces effective stress by 20 % -> strength and bearing capacity fall.

Capillary rise example. Effective grain size D10 = 0.02 mm (silty sand): hc = C/(e D10) with C = 0.3 cm2, e = 0.6 -> hc = 0.3/(0.6 x 0.002) = 250 cm = 2.5 m. Soil in this zone is saturated but with negative pore pressure (suction) -> apparent cohesion.

3.5 Quality of ground water (tests on 1 L sample)

Parameter	Concern	Limit (IS:456 for concrete)
pH	Acidic water attacks concrete	>= 6
Sulphates (as SO4)	Sulphate attack on cement	< 400 mg/l (mix) ; ground: 0.2 % soluble in soil
Chlorides	Corrosion of reinforcement	< 500 mg/l (RCC); 2000 (PCC)
Organic matter, TDS	Setting time, strength	TDS < 2000 mg/l

3.6 Controlling W.T. during construction

- Sump pumping / open pumping - for small seepage in clays & coarse soils.
- Well points - for sands (1 - 6 m), suction lift; deep wells for deep drawdown.
- Cut-off walls (sheet piles, diaphragm walls, grouting) - reduce flow into excavation.
- Freezing or electro-osmosis for silts and very fine soils.
- Always lower W.T. at least 0.5 - 1 m below the excavation level and avoid loss of fines.

4. Preparation of soil profiles

A soil profile (subsoil section) is a vertical section drawn through two or more boreholes showing the sequence of strata, thickness, W.T. and test results. It gives a picture of the ground for design and construction.

4.1 Steps in preparing a profile

- Plot boreholes at horizontal scale (1:200 to 1:500) along the alignment/section line; vertical scale larger (1:50 to 1:100) = exaggerated.
- Reduce depths to R.L. using the ground levels.
- Mark the strata boundary at each borehole and join corresponding layers - using straight/dashed lines. Use dashed lines when the boundary is inferred (uncertain).
- Show pinching out, lenses and faults carefully; do not extend a boundary too far from the boreholes.
- Add graphical symbols, group symbols, N values, W.T. (with date), sample positions, foundation level.

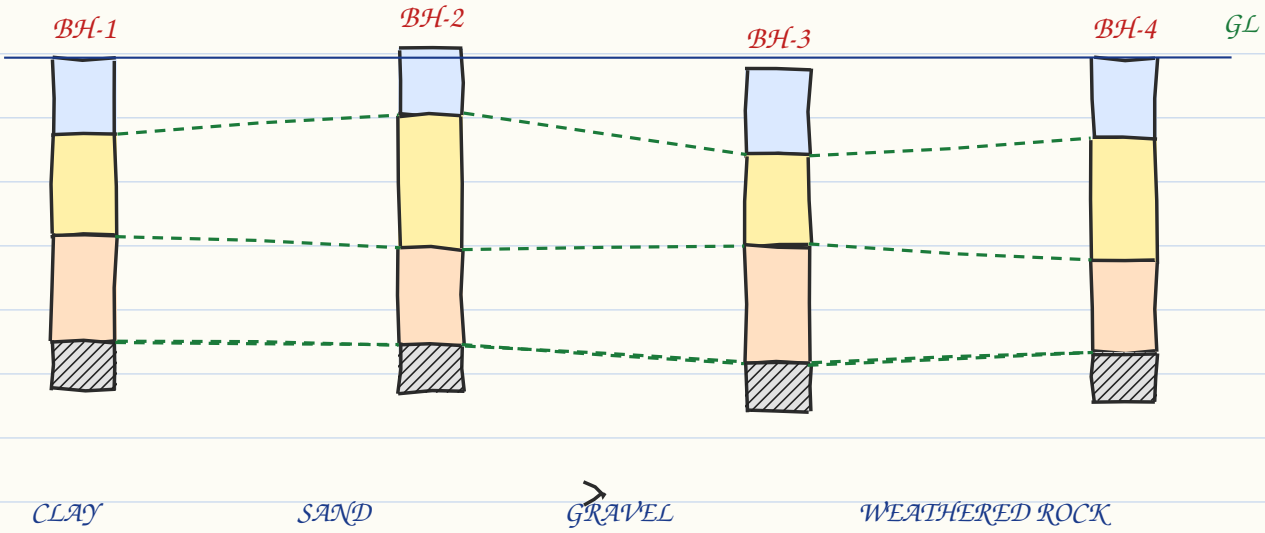


Fig 3.3 Subsoil section through four boreholes

5. Exploration (site investigation) report

The final document of an investigation, containing factual data and the interpretation, with recommendations. Details are in Unit V. Contents in short:

- Introduction: project, location, scope, dates, personnel, investigation method.
- Site description & geology: topography, drainage, regional geology, previous use.
- Field work: boring methods, depth, sampling, tests carried out, borehole logs (annexure).
- Ground water: depth, fluctuation, quality.
- Laboratory tests and results: tabulated.
- Analysis & recommendations: type and depth of foundation, safe bearing capacity / allowable pressure, settlement, special precautions (swelling soil, fill, dewatering).
- Drawings: site plan with borehole location, subsoil sections, graphs.

5.1 Specimen contents of an exploration report

Chapter	Heading
1	Introduction (client, project, objectives, standards followed)
2	Site and geology (location map, topography, drainage, climate, seismic zone)
3	Field investigations (borehole table, methods, in-situ tests, ground water)

Chapter	Heading
4	Laboratory test results (summary sheets, grain size curves, consolidation, shear)
5	Discussion of subsoil conditions (idealised soil profile, soil parameters)
6	Foundation recommendations (type, depth, SBC, settlement)
7	Construction considerations (dewatering, excavation support, special problems)
Annex	Borehole logs, laboratory data sheets, drawings, photographs

Typical recommendation (sample): "Based on the investigation, the site consists of 1.5 m filled soil underlain by stiff silty clay (CI) up to 6 m resting on weathered rock. W.T. was not met up to 12 m (April 2025). Isolated footings at 2.0 m depth below existing G.L. in the stiff clay may be designed for a net safe bearing capacity of 180 kN/m2, limiting settlement to 40 mm."

Do & do not

- Do record actual depths, never "typical" depth; note any change in drilling method.
- Do photograph cores and samples; label boxes.
- Don't leave out disturbed-layers (made-up ground) or boulders.
- Don't mix up borehole numbers or levels - an error can mislead the whole design.

6. Important questions

- Explain how a borehole is logged. Draw a typical borehole log.
- What are the observations to be made regarding ground water in boreholes?
- Explain the effects of water table fluctuations on foundation.
- How is a soil profile prepared from borehole data?
- Write the contents of a soil investigation report.
- Define TCR, SCR and RQD. Compute RQD for a given core run.